# DELHI PUBLIC SCHOOL BULANDSHAHR

Typography
on
API

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XII-F

# CERTIFICATE

*Karan & Vanshika*

This is to certify that Karan & Vanshika of class XII 'F' session 2022-23 has completed her project work entitled ,"API" as a part of the paper of Typography, under my knowledge , it is an excellent piece of work and is worthy of consideration is partial fulfillment of the of the requirement of CBSE for AISSCE practical exam of Typography.

SIGNATURE

**Dr H.S. Vashistha**
**Principal**
**DPS Bulandshahr**

SIGNATURE

**Shekhar Tripathi**
**PGT Typography**
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# ACKNOWLEDGEMENT

Apart from the efforts of me, the success of any project depends largely on the encouragement and guidance of many others. I take this opportunity to express my gratitude to the people who have been instrumental in the successful completion of this project.

I express deep sense of gratitude to almighty god for constant encouragement while carrying out this project.

I gradually acknowledge the contribution of the individuals who contributed in bringing this project up to this level, who continues to look after me despite my flaws,

I express me deep sense of gratitude to the luminary the principal, Ms. H.S. Vashist, Delhi Public School who has been continuously motivating and extending their helping hand to us.

My sincere thanks to Mr. Shekhar Tripathi Master in-change, A guide, Mentor, all the above a friend, who critically reviews during implementation of the project

The guidance and support received from all the members who contributed and who are contributing to this project, was vital for the success of the project. I am grateful for their constant support and help...

# API

**Application Programming Interface**

An Application Programming Interface (API) is a set of functions, procedures, methods or classes used by computer programs to request services from the operating system, software libraries or any other service providers running on the computer. A computer programmer uses the API to make application programs.

API

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# API Brief Overview of Project

An application programming interface (API) is a way for two or more computer programs to communicate with each other. It is a type of software interface, offering a service to other pieces of software. A document or standard that describes how to build or use such a connection or interface is called an API specification. A computer system that meets this standard is said to implement or expose an API. The term API may refer either to the specification or to the implementation. It is not intended to be used directly by a person (the end user) other than a computer programmer who is incorporating it into the software. An API is often made up of different parts which act as tools or services that are available to the programmer. A program or a programmer that uses one of these parts is said to call that portion of the API. The calls that make up the API are also known as subroutines, methods, requests, or endpoints. An API specification defines these calls, meaning that it explains how to use or implement them

# API

## Purpose

In building applications, an API simplifies programming by abstracting the underlying implementation and only exposing objects or actions the developer needs. While a graphical interface for an email client might provide a user with a button that performs all the steps for fetching and highlighting new emails, an API for file input/output might give the developer a function that copies a file from one location to another without requiring that the developer understand the file system operations occurring behind the scenes.

# API

## History

**1968**

The idea of the API is much older than the term itself. British computer scientists Maurice Wilkes and David Wheeler worked on a modular software library in the 1940s for EDSAC, an early computer..

**1968**

The term "application program interface" is first recorded in a paper called Data structures and techniques for remote computer graphics.

# API

## History

**1990**

By 1990, the API was defined simply as "a set of services available to a programmer for performing certain tasks" by technologist Carl Malamud.

**2000**

Roy Fielding's dissertation Architectural Styles and the Design of Network-based Software Architectures at UC Irvine in 2000 outlined Representational state transfer (REST) and described the idea of a "network-based Application Programming Interface" that Fielding contrasted with traditional "library-based" APIs

# API

## Usage

### WEB API

***Web APIs are a service accessed from client devices (Mobile Phones, Laptop, etc.) to a web server using the Hypertext Transfer Protocol (HTTP)***

### REMOTE API

***Remote APIs allow developers to manipulate remote resources through protocols, specific standards for communication that allow different technologies to work together, regardless of language or platform.***

### OPERATING SYSTEM

*An API can specify the interface between an application and the operating system. POSIX, for example, specifies a set of common APIs that aim to enable an application written for a POSIX conformant operating system to be compiled for another POSIX conformant operating system.*

API

# Design

The design of an API has a significant impact on its usage. First of all, the design of programming interfaces represents an important part of software architecture, the organization of a complex piece of software.

API

# Security

API security is very critical when developing a public facing API. Common threats include SQL injection, Denial-of-service attack (DoS), broken authentication, and exposing sensitive data.

# API

## Release policies

### Partner

Only specific business partners can use the API

### Private API

The API is for internal company use only.

### Public

The API is available for use by the public

# API

## The API is available for use by the public

An important factor when an API becomes public is its "interface stability". Changes to the API—for example adding new parameters to a function call—could break compatibility with the clients that depend on that API.

When parts of a publicly presented API are subject to change and thus not stable, such parts of a particular API should be documented explicitly as "unstable“.

A public API can sometimes declare parts of itself as deprecated or rescinded. This usually means that part of the API should be considered a candidate for being removed, or modified in a backward incompatible way. Therefore, these changes allow developers to transition away from parts of the API that will be removed or not supported in the future

# API

## Documentation

1. API documentation describes the services an API offers and how to use those services, aiming to cover everything a client would need to know for practical purposes.
2. Documentation is crucial for the development and maintenance of applications using the API.API documentation is traditionally found in documentation files but can also be found in social media such as blogs, forums, and Q&A websites.
3. Traditional documentation files are often presented via a documentation system, such as Javadoc or Pydoc, that has a consistent appearance and structure. However, the types of content included in the documentation differ from API to API.
4. In the interest of clarity, API documentation may include a description of classes and methods in the API as well as "typical usage scenarios, code snippets, design rationales, performance discussions, and contracts", but implementation details of the API services themselves are usually omitted.

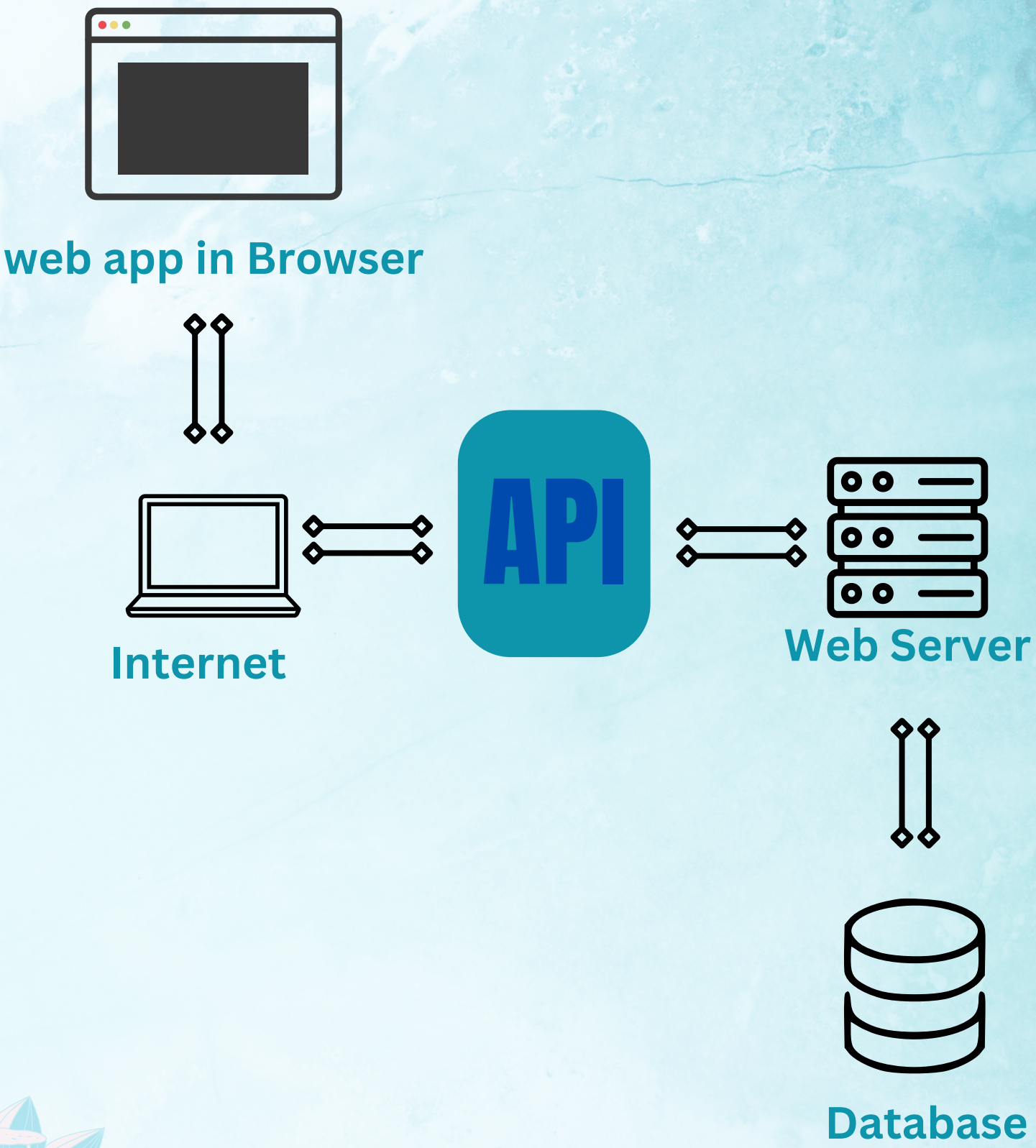
# API

## Dispute over copyright protection for API

In 2010, Oracle Corporation sued Google for having distributed a new implementation of Java embedded in the Android operating system. Google had not acquired any permission to reproduce the Java API, although permission had been given to the similar Open JDK project. Google had approached Oracle to negotiate a license for their API, but were turned down due to trust issues. Despite the disagreement, Google chose to use Oracle's code anyway. Judge William Alsup ruled in the *Oracle v. Googel* case that APIs cannot be copyrighted in the U.S and that a victory for Oracle would have widely expanded copyright protection to a "functional set of symbols" and allowed the copyrighting of simple software commands

# API

## How API Works.



API

# Some API tools

- Katalon Studio
- TestGrid
- ReadyAPI
- Subject7
- Eggplant
- Postman

API

# KATALON STUDIO

**Features**:

:- Supports all types of REST, SOAP requests, SSL client certificate.

:- Easily import tests from Swagger (2.0 & 3.0), Postman, WSDL, and WADL.

:- Built off of data-driven testing methods better test coverage and reliability.

:- Create fluent assertions in BDD style with AssertJ support.

:- Support set up API test data using UI testing.

:- Provide insightful test report dashboards of all testing stages for better monitor and collaboration across teams.

# API

## EGGPLANT

**Features**:

- Understand how the API behaves with both valid and invalid values.
- Easily add additional instructions and steps beyond simple API call and response.
- Discover and pinpoint anomalous behavior before your users encounter it.
- Auto-generate test automation assets by collecting and analyzing metadata by hooking into APIs.
- Combine API testing with validation of the UI to understand real-world performance and behavior.
- Easily build and execute API tests with pre-configured snippets, parameters, and SenseTalk scripts.

API

SMARTBEAR

# READY API

**Features**:

- ReadyAPI can be integrated into any environment.
- It has a Smart Assertion feature that can create bulk assertions against hundreds of endpoints quickly.
- Native support for Git, Docker, Jenkins, Azure, etc.
- Supports Command-line for automated testing.
- Supports parallel execution of functional tests and job queuing.
- Promotes code reuse
- Removes dependencies during testing and development.

# API

# TESTGRID

**Features**:

- Easy to Use API Automation for SOAP and REST calls
- Deeper Assertions – has the ability to write test cases that validate against API calls. Supports MITM
- Fetch detailed report of the number of Source Type, Request Data, Response Data, Duration, Speed, Request Size, and Response Size.
- Performance testing to help you optimize & improve your API
- API Integration Automated Testing
- Transaction Monitoring
- Scriptless testing for faster automation
- Integrate with your favorite CI/CD tool for continuous testing

# API

SUBJECT7s
QUALITY IS A CULTURE

**Key Features:**

- Supports both REST and SOAP services
- Enables verification of responses in JSON and XML formats
- Reads and stores values from API response, easily incorporating them into the testing flow.
- Simple, non-metered pricing, delivering financial predictability.
- SOC2 Type2 compliant

# API

POSTMAN

**Features:**

- With Postman, almost all modern web API data can be extracted
- You can write Boolean tests within Postman Interface
- You can create a collection of REST calls and save each call as part of a collection for execution in future
- Unlike CURL, it is not a command line based tool, which makes this tool hassle free of pasting text into command line window
- For transmitting and receiving REST information, Postman is more reliable